

EXPERIMENT – 6

Snort IDS/IPS to Detect and Prevent Real-Time Threats in TryHackMe Platform

AIM

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE / TOOLS REQUIRED

TryHackMe account
AttackBox or VPN connection
Snort IDS/IPS
Nmap
Linux Terminal

PROCEDURE

1. Starting TryHackMe Environment
2. Login to TryHackMe.
3. Search for the Snort room.
4. Click Start Machine.
5. Click Start AttackBox.
Open Terminal in AttackBox.
6. Verify Snort Installation
snort -V
7. Test Snort Configuration
sudo snort -T -c /etc/snort/snort.conf

EXPERIMENT 1: ICMP (Ping) Detection – IDS Mode

Edit Local Rules

```
sudo nano /etc/snort/rules/local.rules
```

Add Rule

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Save and Exit

Ctrl + O → Enter

Ctrl + X

Run Snort in IDS Mode

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

Generate ICMP Traffic (New Terminal Tab)

```
ping 8.8.8.8
```

Observation

Alert “ICMP Ping Detected” is displayed in Snort terminal.

(Optional log check)

```
sudo tail -f /var/log/snort/alert
```

EXPERIMENT 2: Port Scan Detection

Edit Rules File

```
sudo nano /etc/snort/rules/local.rules
```

Add Rule

```
alert tcp any any -> any any (flags:S; msg:"Port Scan Detected"; sid:1000002; rev:1;)
```

Restart Snort

Ctrl + C

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

Run Nmap Scan (New Terminal)

```
nmap -sS 8.8.8.8
```

Observation

Snort detects the SYN scan and displays "Port Scan Detected" alert.

EXPERIMENT 3: IPS Mode – Blocking ICMP (Inline Mode)

Modify Rules

```
sudo nano /etc/snort/rules/local.rules
```

Add Rule

```
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)
```

Run Snort in Inline (IPS) Mode

```
sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf
```

Generate ICMP Traffic (New Terminal)

```
ping 8.8.8.8
```

(Optional)

```
sudo tail -f /var/log/snort/alert
```

NOTE

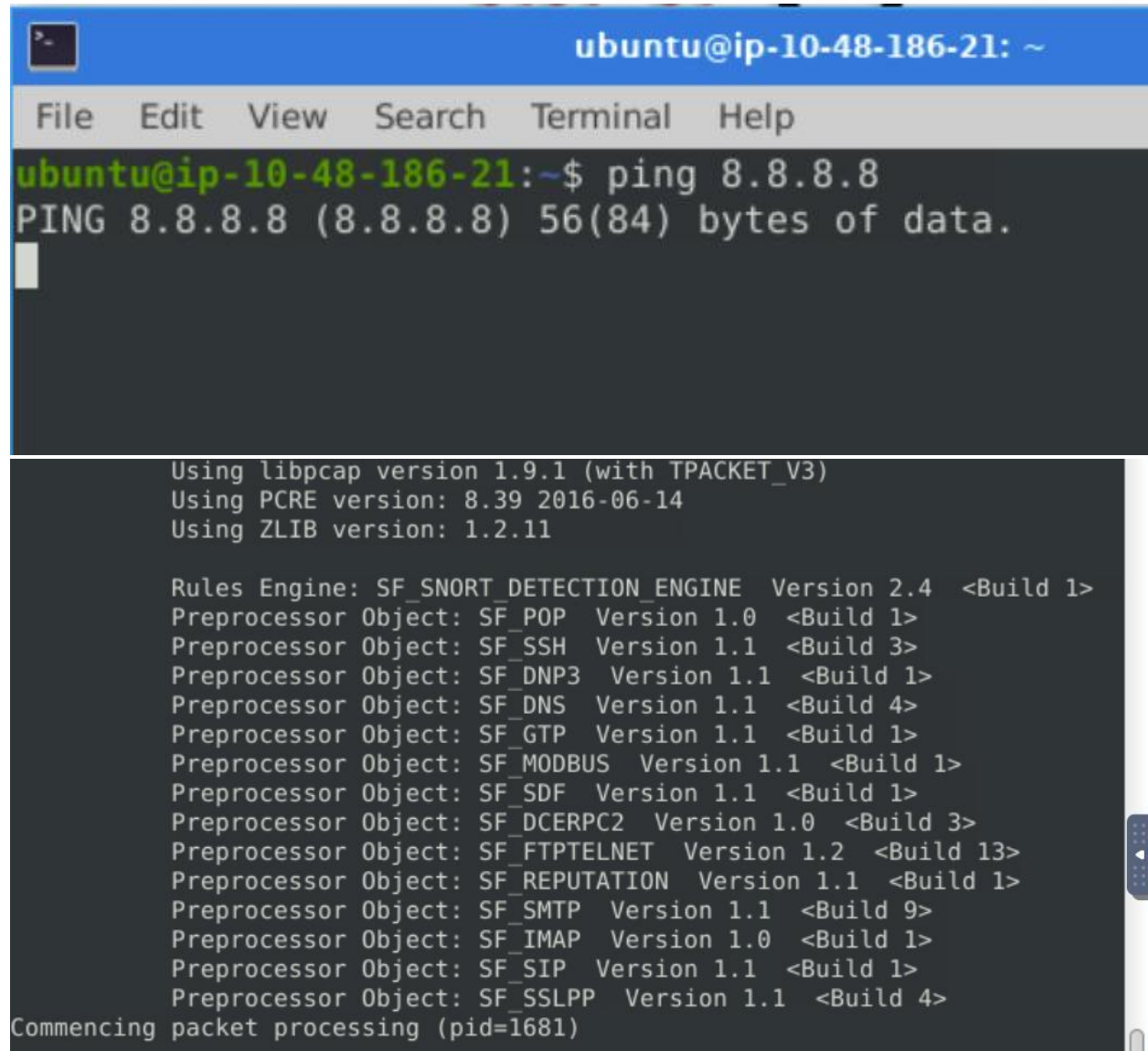
If eth0 does not work:

```
ip a
```

Use the correct interface name instead of eth0.

OUTPUT

Snort successfully detected ICMP traffic, identified port scanning activity, and blocked ICMP packets in inline IPS mode.

A screenshot of a terminal window titled 'ubuntu@ip-10-48-186-21: ~'. The terminal has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The user has entered the command 'ping 8.8.8.8'. The output shows 'PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.' followed by a blank line. Below this, the terminal displays the Snort configuration details, including the version of libpcap, PCRE, and ZLIB, followed by a list of preprocessors and their versions. The final line indicates that packet processing has commenced with PID 1681.

```
ubuntu@ip-10-48-186-21: ~  
File Edit View Search Terminal Help  
ubuntu@ip-10-48-186-21:~$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
  
Using libpcap version 1.9.1 (with TPACKET_V3)  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.2.11  
  
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 2.4 <Build 1>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>  
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>  
Commencing packet processing (pid=1681)
```

RESULT

Thus, Snort IDS/IPS was successfully configured on the TryHackMe platform to detect and prevent real-time network threats.